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De Rham Theory

1.1 De Rham Cohomology

Definition 1.1.1

The *q-th de Rham cohomology* of $\mathbb{R}^n$ is the vector space

$$H_{DR}^q(\mathbb{R}^n) = \{\text{closed } q\text{-forms}\} / \{\text{exact } q\text{-forms}\}.$$

We sometimes suppress the subscript *DR* and write $H^q(\mathbb{R}^n)$. If there is a need to distinguish between a form ω and its cohomology class, we denote the latter by $[\omega]$.

Remark.

The definition works equally well for any open subset $U \subseteq \mathbb{R}^n$. So we may also speak of the de Rham cohomology $H_{DR}^*(U)$ of U .

Example 1.1.2

1. When $n = 0$,

$$H^q = \begin{cases} \mathbb{R}, & q = 0 \\ 0, & q > 0 \end{cases}$$

2. When $n = 1$. $(\ker d) \cap \Omega^0(\mathbb{R}^1)$ are the constant functions

$$H^0(\mathbb{R}^1) = \mathbb{R}$$

On $\Omega^1(\mathbb{R}^1)$, $\ker d$ are all the 1-forms. If $\omega = g(x) dx$ is a 1-form, then by taking

$$f = \int_0^x g(u) du,$$

we find that

$$df = g(x) dx$$

Therefore every 1-form on $\mathbb{R}^1$ is exact and

$$H^1(\mathbb{R}^1) = 0.$$

3. Let U be a disjoint union of m open intervals on $\mathbb{R}^1$. Then

$$H^0(U) = \mathbb{R}^m$$

and

$$H^1(U) = 0$$

4. In general

$$H^*(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & \text{in dimension 0,} \\ 0, & \text{otherwise} \end{cases}$$

This result is called the Poincare lemma and will be proved later.

Definition 1.1.3

Let $\Omega_c^*(\mathbb{R}^n)$ be the collection of the compact supported forms on $\mathbb{R}^n$, which is a cochain complex. Then we denote the cohomology of this complex by $H_c^*(\mathbb{R}^n)$

Example 1.1.4

1.

$$H_c^*(\text{point}) = \begin{cases} \mathbb{R}, & \text{in dimension 0,} \\ 0, & \text{elsewhere.} \end{cases}$$

2. The compact cohomology of $\mathbb{R}^1$. Again the closed 0-forms are the constant functions. Since there are no constant functions on $\mathbb{R}^1$ with compact support.

$$H_c^0(\mathbb{R}^1) = 0.$$

To compute $H_c^1(\mathbb{R}^1)$, consider the integration map

$$\int_{\mathbb{R}^1} : \Omega_c^1(\mathbb{R}^1) \rightarrow \mathbb{R}$$

This map is clearly surjective. It vanishes on the exact 1-forms df where f has compact support, for if the support of f lies in the inte-

rior of $[a, b]$, then

$$\int_{\mathbb{R}^1} \frac{df}{dx} dx = \int_a^b \frac{df}{dx} dx = f(b) - f(a) = 0$$

If $g(x) dx \in \Omega_c^1(\mathbb{R}^1)$ is in the kernel of the integration map, then the function

$$f(x) = \int_{-\infty}^x g(u) du$$

will have compact support and $df = g(x) dx$. Hence the kernel of $\int_{\mathbb{R}^1}$ are precisely the exact forms and

$$H_c^1(\mathbb{R}^1) = \frac{\Omega_c^1(\mathbb{R}^1)}{\ker \int_{\mathbb{R}^1}} = \text{Im } \int_{\mathbb{R}^1} = \mathbb{R}^1$$

3. More generally,

$$H_c^*(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & \text{in dimension } n \\ 0, & \text{otherwise.} \end{cases}$$

This result is the Poincare lemma for cohomology with compact support and will be proved later.

Exercise 1.1.5

Compute $H_{DR}^*(\mathbb{R}^2 - P - Q)$ where P and Q are two points in $\mathbb{R}^2$. Find the closed forms that represent the cohomology classes.

Proof. First ,

$$H_{DR}^k(\mathbb{R}^2 - P - Q) = 0, \quad k \geq 3, \quad H_{DR}^0(\mathbb{R}^2 - P - Q) \simeq \mathbb{R}.$$

Since there is no nonzero 3-form on $\mathbb{R}^2 - P - Q$. To compute $H_{DR}^2(\mathbb{R}^2 - P - Q)$. Consider a 2-form on $\alpha = f(x, y) dx dy$ on $\mathbb{R}^2 - P - Q$. Since $\mathbb{R}^2 - P - Q$ is open, it is a submanifold of $\mathbb{R}^2$. It follows that α is closed from Stokes Theorem.

Finally, let $\omega = g(x, y) dx + h(x, y) dy$. Then

$$d\omega = (g_x - h_y) dx \wedge dy$$

We aim to find a solution of

$$g_x - h_y = f$$

on $\mathbb{R}^2 - P - Q$. It actually exists. Let α be any closed 1-form on $\mathbb{R}^2 - P - Q$. We may assume that $P = (-1, 0)$, $Q = (1, 0)$. Let

$$\omega_P = \frac{-y dx + (x+1) dy}{(x+1)^2 + y^2}, \quad \omega_Q = \frac{-y dx + (x-1) dy}{(x-1)^2 + y^2}$$

Then

$$d\omega_P = d\omega_Q = 0,$$

Let C_P, C_Q be two sufficiently small loops around P, Q , respectively. And C_P, C_Q is not around Q, P , respectively. Then

$$\int_{C_P} \omega_P = \int_{C_Q} \omega_Q = 2\pi, \quad \int_{C_Q} \omega_P = \int_{C_P} \omega_Q = 0$$

For any 1-form α , we define

$$c_P(\alpha) = \frac{1}{2\pi} \int_{C_P} \alpha, \quad c_Q(\alpha) = \frac{1}{2\pi} \int_{C_Q} \alpha$$

For any loops on $\mathbb{R}^2 - P - Q$, it is homotopic to a linear combination of C_P and C_Q . Let

$$\beta = \alpha - c_P(\alpha) \omega_P - c_Q(\alpha) \omega_Q$$

Since any loop γ , $\mathbb{R}^2 - P - Q$ is homotopic to a combination of C_P and C_Q . Thus

$$\int_{\gamma} \beta \equiv 0$$

Thus β is an exact form. That is to say

$$[\alpha] = c_P(\alpha) [\omega_P] + c_Q(\alpha) [\omega_Q]$$

Thus

$$H_{DR}^1(\mathbb{R}^2 - P - Q) = \text{span} \langle [\omega_P], [\omega_Q] \rangle \simeq \mathbb{R}^2$$

And the represent classes of α is $c_P(\alpha) [\omega_P] + c_Q(\alpha) [\omega_Q]$ □

1.2 The Mayer-Vietoris Sequence

Definition 1.2.1

Let $x_1, \dots, x_m$ and $y_1, \dots, y_n$ be the standard coordinates on $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively. Let $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a smooth map. The commutativity with d defines the pullback map f^* uniquely :

$$f^* \left(\sum g_I dy_{i_1} \cdots dy_{i_q} \right) = \sum (g_I \circ f) df_{i_1} \cdots df_{i_q},$$

where $f_i = y_i \circ f$ is the i -th component of the function f .

Proposition 1.2.2

f^* commutes with d

Proof.

$$df^* (g_I dy_{i_1} \cdots dy_{i_q}) = d((g_I \circ f) df_{i_1} \cdots df_{i_q}) = d(g_I \circ f) df_{i_1} \cdots df_{i_q}$$

$$\begin{aligned} f^* d(g_I dy_{i_1} \cdots dy_{i_q}) &= f^* \left(\sum_i \frac{\partial g_I}{\partial y_i} dy_i dy_{i_1} \cdots dy_{i_q} \right) \\ &= \sum_i \left(\frac{\partial g_I}{\partial y_i} df_i \right) df_{i_1} \cdots df_{i_q} \\ &= d(g_I \circ f) df_{i_1} \cdots df_{i_q} \end{aligned}$$

□

Proposition 1.2.3

Let g be a smooth function on $\mathbb{R}^n$. dg is independent of the coordinate system.

Proposition 1.2.4

If $\omega = \sum g_I du_I$, then $d\omega = \sum dg_I du_I$

Proposition 1.2.5

Ω^* is a **contravariant functor** from the category of **Euclidean spaces** $\{\mathbb{R}^n\}_{n \in \mathbb{Z}}$ and **smooth maps**: $\mathbb{R}^m \rightarrow \mathbb{R}^n$ to the category of **commutative differential graded algebras** and their **homomorphisms**. It is the **unique** such functor that is the **pullback of functions** on $\Omega^0(\mathbb{R}^n)$. Here the commutativity of the graded algebra refers to the fact that

$$\tau\omega = (-1)^{\deg \tau \deg \omega} \omega\tau.$$

Remark.

The functor Ω^* may be extended to the category of differential manifolds.

Theorem 1.2.6 ▶ Mayer-Vietoris Sequence

Suppose $M = U \cup V$ with U, V open. Then there is a sequence of inclusions

$$M \longleftarrow U \amalg V \begin{matrix} \xleftarrow{\partial_1} \\ \xleftarrow{\partial_0} \end{matrix} U \cap V$$

where $U \amalg V$ is the disjoint union of U and V and ∂_0 and ∂_1 are the inclusions of $U \cap V$ in V and in U respectively. Applying the contravariant functor Ω^* , we get a sequence of restrictions of forms

$$\Omega^*(M) \longrightarrow \Omega^*(U) \oplus \Omega^*(V) \begin{matrix} \xrightarrow{\partial_0^*} \\ \xrightarrow{\partial_1^*} \end{matrix} \Omega^*(U \cap V),$$

where by the restriction of a form to a submanifold we mean its image under the pullback map induced by the inclusion. By taking the difference of the last two maps, we obtain the **Mayer-Vietoris sequence**

$$0 \longrightarrow \Omega^*(M) \longrightarrow \Omega^*(U) \oplus \Omega^*(V) \longrightarrow \Omega^*(U \cap V) \longrightarrow 0$$

$$(\omega, \tau) \longmapsto \tau - \omega$$

Proposition 1.2.7

The Mayer-Vietoris sequence is exact.

Proof.

□